DARK PATTERNS POST-GDPR

Scraping Consent Interface Designs and Demonstrating Their Influence

Nouwens, Liccardi, Veale, Karger & Kagal (CHI 2020)

WHAT PROBLEM?	WHY IMPORTANT?	WHAT DID THEY FIND?
Cookie banners may record consent without giving users a genuinely free and informed choice.	Tracking can expose personal data to hundreds of third parties, affecting autonomy, privacy and legal compliance.	Interface design strongly changes consent rates; many post-GDPR banners remain minimally non-compliant.

1. Title

The title signals a two-part investigation: the researchers first scrape real cookie-consent interfaces used after the GDPR, then experimentally test whether those designs influence users. "Dark patterns" means interface choices that steer users toward the company-preferred action, usually accepting tracking.

2. Abstract

The paper examines Consent Management Platforms (CMPs), the systems that display cookie and privacy notices. It combines a large-scale website study with a controlled user experiment.

- Study 1 scraped 680 implementations of the five most common CMPs found among the top 10,000 UK websites.
- Study 2 tested eight interface designs with 40 participants during normal web browsing.
- Only 11.7% of sites met the authors' three minimum GDPR-based conditions.
- Hiding "Reject All" beyond the first screen increased consent by 22-23 percentage points.
- Showing more detailed purposes or vendors on the first screen reduced consent by 8-20 percentage points.

3. Introduction: first 1-2 paragraphs

The web mainly uses a "notice and consent" model: a website presents information, then obtains or assumes agreement to collect, store and process personal data. In practice, these notices appear as cookie banners.

The central problem is that legal consent must be freely given, specific, informed, unambiguous and based on an affirmative action. Yet many interfaces make acceptance quick and rejection difficult, or treat scrolling, continued browsing or closing the banner as consent. This raises a basic question: is the user choosing, or is the interface choosing for the user?

4. Conclusion

The authors conclude that non-compliant or manipulative designs remain widespread after the GDPR. Information and controls hidden below the first layer are usually ignored, so apparent consent often fails to represent genuine privacy preferences.

- Regulators should use automated auditing tools and enforce rules against CMP vendors, not only individual websites.
- CMPs should not permit configurations that use implied consent, pre-ticked options or unequal accept/reject paths.
- Long-term solutions may require durable browser-level privacy settings rather than repeated site-by-site banners.

RESEARCH DESIGN, FINDINGS & ETHICAL IMPLICATIONS

5. Research Questions

- What is the current state of CMP interface design in the EU, and how common are non-compliant elements?
- How do interface designs affect users' consent actions and whether consent is truly "freely given"?

6. Methodology

STUDY 1 - WEB SCRAPING	STUDY 2 - FIELD EXPERIMENT
• Scraped the top 10,000 UK websites; 680 usable CMP implementations. • Covered QuantCast, OneTrust, TrustArc, Cookiebot and Crownpeak. • Tested three minimum conditions: explicit consent, equal ease of accept/reject, and no pre-ticked optional choices.	• 40 participants used a Chrome extension that injected eight consent interfaces. • Compared banner vs barrier, visible vs hidden rejection, and bulk vs granular choices. • Recorded behaviour and analysed design effects with regression models; followed by a post-study survey.

7. Results

WEBSITE FINDINGS	USER EXPERIMENT
• 32.5% used implicit consent. • 50.1% had no "Reject All" button. • Only 12.6% made rejection as easy as acceptance. • 56.2% pre-ticked optional purposes or vendors. • Median number of listed third parties: 315. • Only 11.7% met all three minimum conditions.	• Banner vs barrier did not materially change acceptance. • Removing first-page rejection raised consent by 22-23 points. • Granular first-page controls lowered consent by 8-20 points. • 93.1% of interactions stayed on the first page. • Only 1.3% of responses were genuinely personalised. • Many users said actual choices did not match ideal preferences.

8. Discussion

The evidence shows that consent is partly an interface-produced outcome, not simply a stable personal preference. When rejection is hidden or requires extra effort, more users accept. When consequences are made visible through granular options, acceptance falls. The first screen therefore carries most of the ethical and legal weight.

Key limitation: the participant sample was small, young, mostly university-educated and based in the United States. The exact percentages should not be treated as universal, but the direction of the design effect is clear.

Applied Answers for Web Application Design

How much data?	Collect only what is necessary for a defined function; minimise scope, retention and third-party sharing.
Is consent enough?	No. Ethical collection also requires necessity, fairness, transparency, security, purpose limitation and accountability.
Effect of dark patterns?	They exploit friction, urgency and fatigue, causing users to accept choices that may not reflect their real preferences.
Engineer responsibility?	Design equal accept/reject paths, no preselected tracking, no collection before consent, easy withdrawal and accurate enforcement.
Privacy vs personalization?	Offer a useful basic service without tracking; request optional personalization separately and explain the concrete benefit.

Bottom line: A consent banner is ethical only when it preserves user autonomy, not merely when it records a click.

Source: Nouwens et al. (2020), "Dark Patterns Post-GDPR: Scraping Consent Interface Designs and Demonstrating Their Influence," CHI 2020.